# Info About AI Usage

I put colored dots on every right corner to show how i used ai on every step.

Image
Genartation

Idea
Genaration

Reasearch

# MILESTONE 1
# Empathize

## KEY RESEARCH FINDINGS

- Traditional Turkish tea systems are designed for multiple people
- People living alone do not want to brew a full teapot
- Tea bags are practical but lack the flavor of loose Turkish tea and a source of microplastics

One of the main findings is that traditional loose-leaf tea provides better taste and a more authentic experience, but it requires more time and effort to prepare. In contrast, tea bags are widely preferred for their speed and practicality. However, many tea bags contain plastic components, which can release microplastics into hot water, creating environmental and health concerns.

Traditional loose-leaf tea provides better taste and aroma but requires more time and preparation.

- Tea bags are preferred because they brew quickly due to finely ground tea particles and permeable packaging.
- Many commercial tea bags contain plastic components that may release microplastics when exposed to hot water.

A single plastic tea bag brewed at 95 °C releases about 11.6 billion microplastic particles and 3.1 billion nanoplastic particles into one cup of tea.

# MILESTONE 2
# Define

Traditional Turkish tea brewing systems are designed for multiple servings and are not practical for individuals living alone. While tea bags offer convenience, they compromise flavor and may release microplastics. This creates an opportunity to design a single-serve brewing device that preserves the authentic taste of loose-leaf Turkish tea while simplifying the preparation process.

## PROBLEM FRAMING

Current tea preparation methods force users to choose between convenience and quality. Tea bags are fast but create waste and may contain microplastics, while loose-leaf tea offers better taste but requires more time and tools.

## DESIGN GOAL

The goal of the project is to design a device that:

- Brews authentic Turkish tea
- Works for single-serve preparation
- Combines multiple functions in one object

The product should simplify the tea preparation process while maintaining the traditional flavor experience.

## OPPORTUNITY AREAS

- Developing a system that combines loose-leaf quality with tea bag speed.
- Eliminating plastic and paper tea bags to reduce waste and microplastic exposure.
- Creating a reusable, integrated brewing solution with minimal user effort.
- Designing a compact, user-friendly product suitable for small living spaces.

# MILESTONE 3
# Ideate

## 1. CONVENIENCE

- Minimal preparation steps
- Reduced number of separate tools
- Easy cleaning
- Compact storage

## 2. USER EXPERIENCE

- Intuitive operation
- Clear brewing logic
- Sensory satisfaction (grinding ritual, aroma release)

## 3. FEASIBILITY

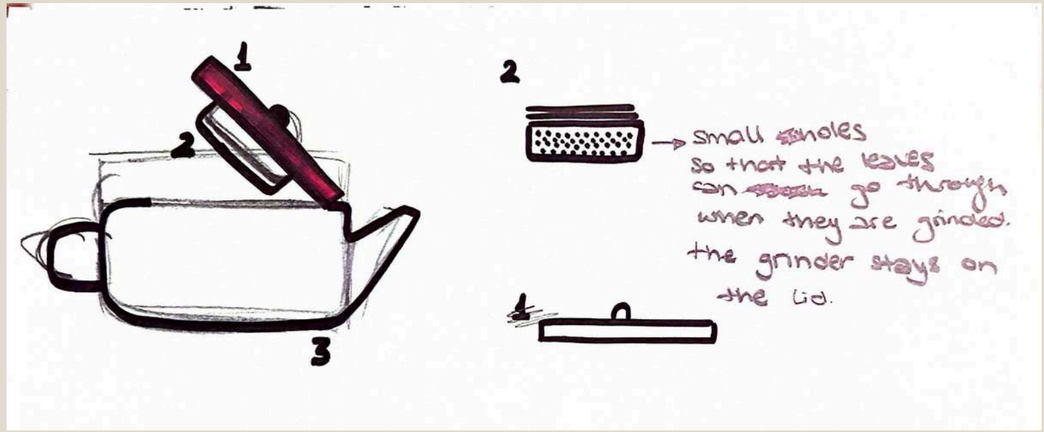
- Mechanically achievable
- Reasonable production complexity
- Suitable for kitchen context

## CONCEPT DEVELOPMENT

During concept exploration, several ideas were considered. The concept evolved around integrating:

- a tea grinding mechanism
- a brewing chamber
- a filtering system

The grinder breaks the tea leaves into smaller pieces, allowing faster extraction and stronger flavor.
A French press style filter was explored as a way to trap the small tea particles after brewing.

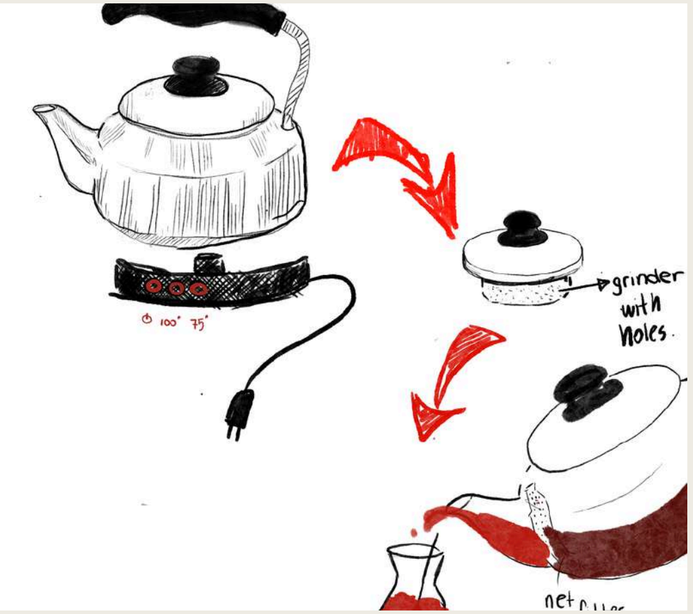


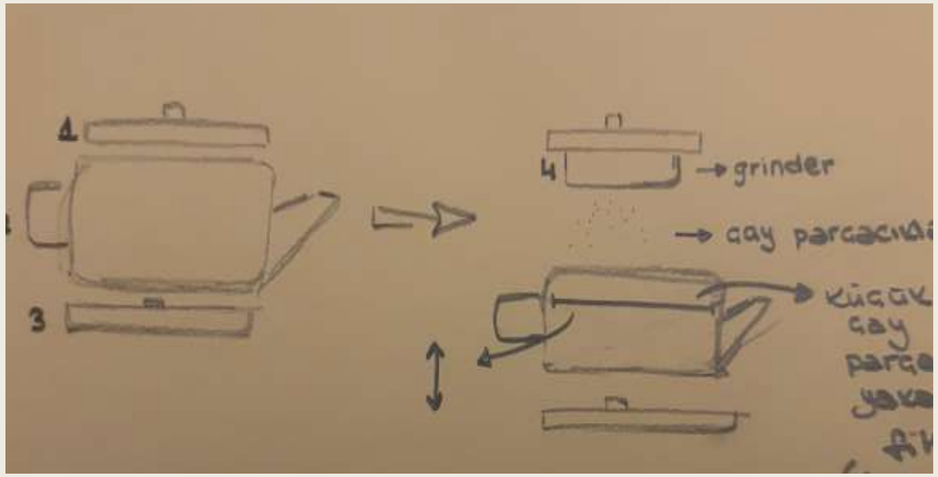
# MILESTONE 3

# Ideate

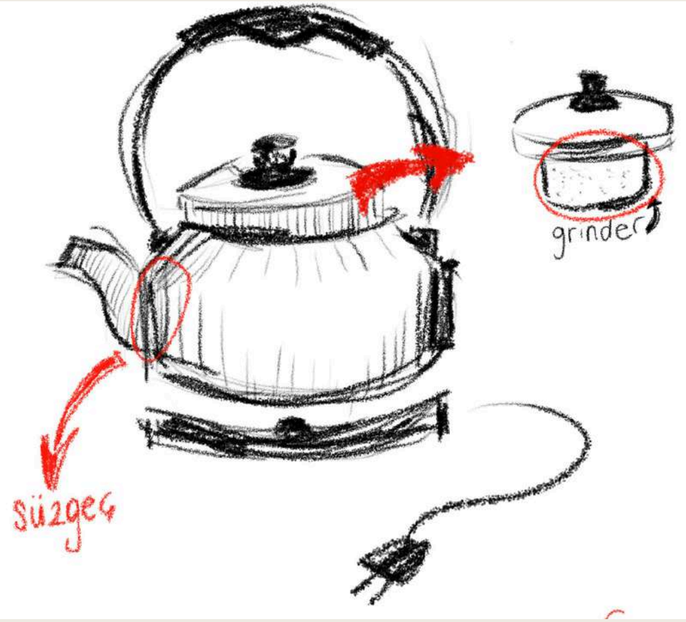
## REJECTED IDEAS

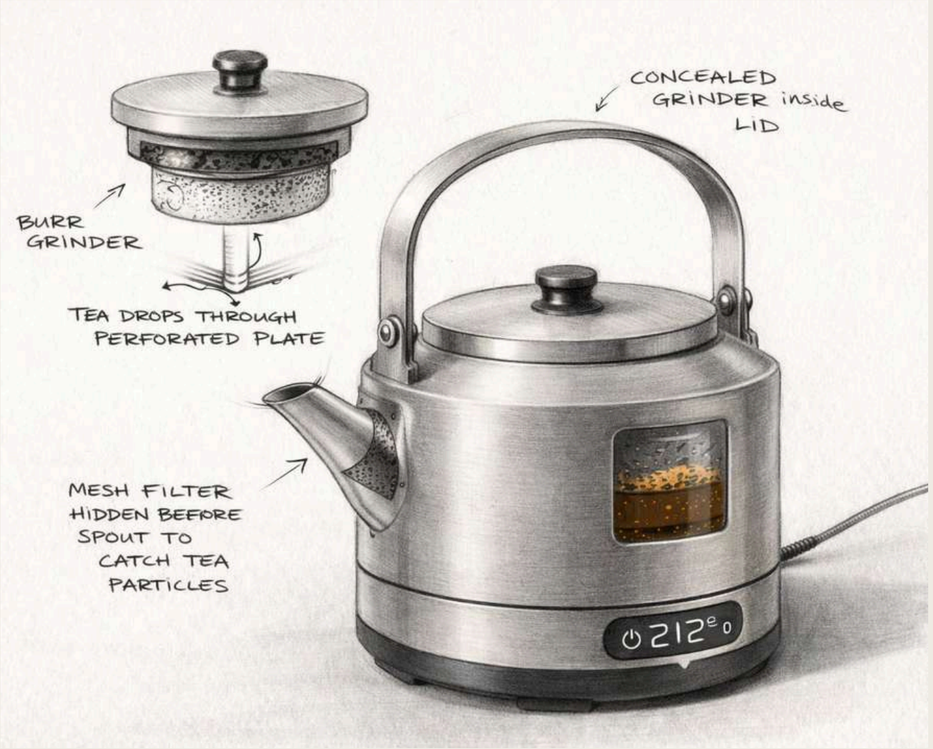
I rejected these designs because I wanted to avoid a form that resembles a coffee maker or a French press. Instead of designing the grinder mechanism as a separate component, I chose to integrate it into the overall body of the machine, making it part of the product as a whole.


100° 75°
grinder with holes.


4
3
4
grinder

212°


grinder
süzgeç


CONCEALED GRINDER inside LID
BURR GRINDER
TEA DROPS THROUGH PERFORATED PLATE
MESH FILTER HIDDEN BEFORE SPOUT TO CATCH TEA PARTICLES
212°

212°

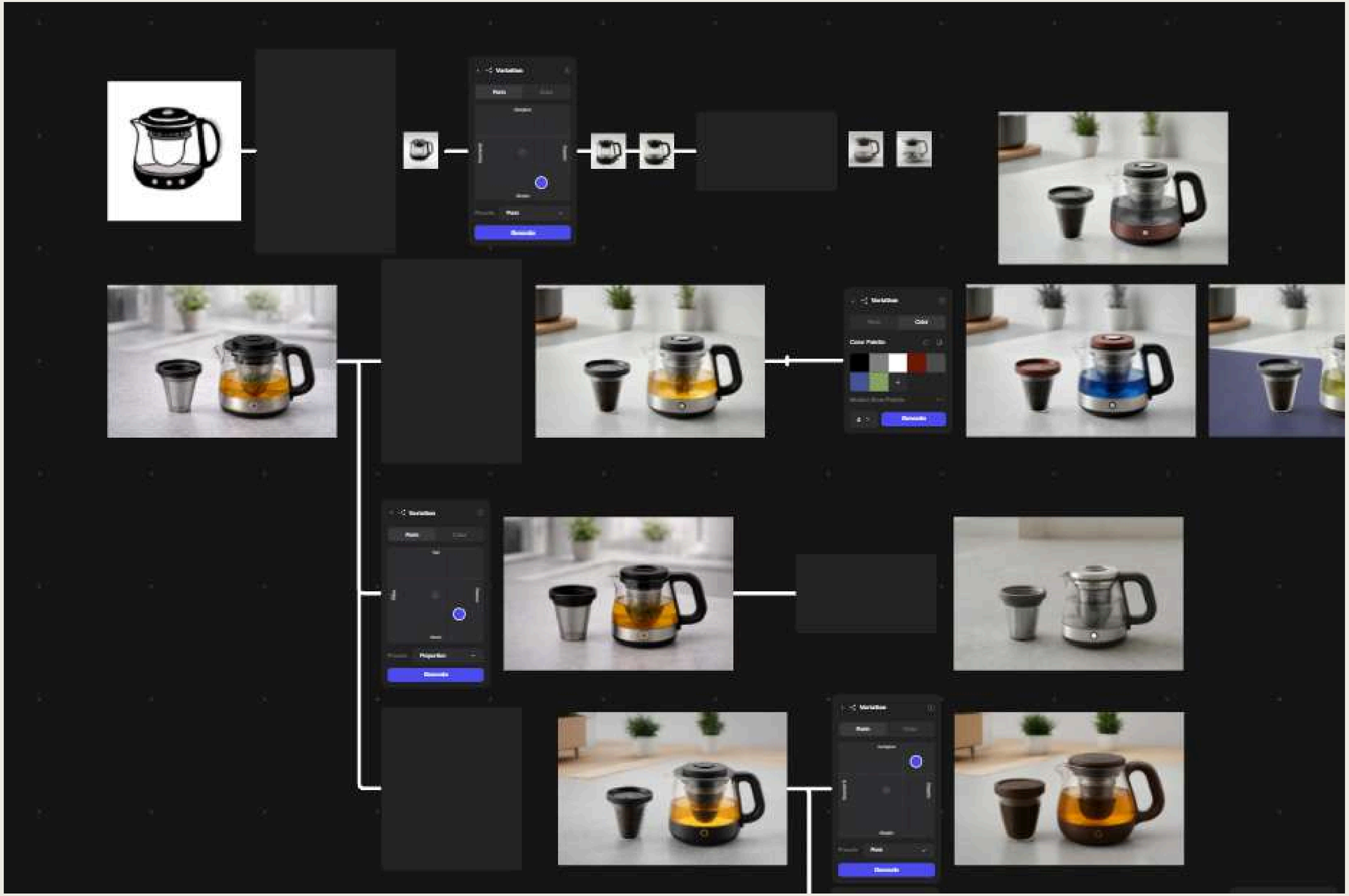
# MILESTONE 3
# Ideate

## MECHANISM

The proposed mechanism works in several steps:

- Tea leaves are placed in the grinder chamber.
- The grinder breaks the leaves into smaller particles.
- The tea mixes with hot water inside the brewing chamber.
- After steeping, a press filter moves downward.
- The filter traps the small tea particles at the bottom.
- Clear tea can then be poured from the spout.

This system combines grinding, brewing, and filtering in a single device.


Variation
Color
Color Palette
Variation
Proportion
Variation
Pose

# MILESTONE 4
## Final Design